

EXCEL DATA MODELLING PROJECT

Project overview:

This project demonstrates the ability to transform raw, fragmented data into a structured relational model. Using excel data modelling project.xlsx, I established complex connections between multiple tables to facilitate deep-dive analysis and automated reporting.

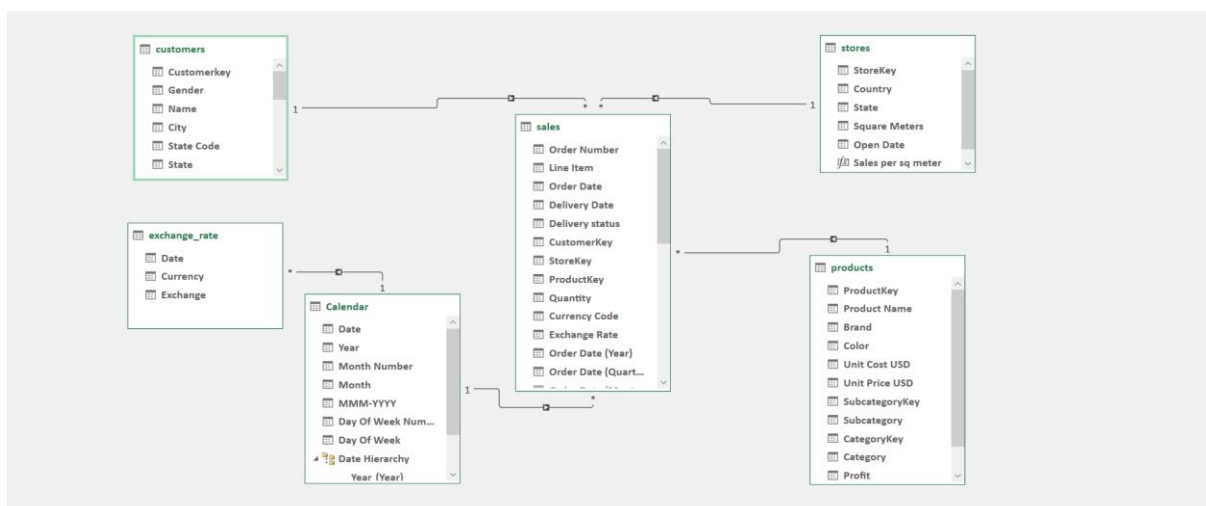
Key Features

- **Relational Data Modelling:** Created a "Star Schema" by linking dimension tables to fact tables using Power Pivot.
- **DAX Measures:** Authored custom Data Analysis Expressions (DAX) to calculate advanced metrics and KPIs.
- **Data Integrity:** Implemented data cleaning and normalization techniques to ensure accuracy across 8 integrated sheets.
- **Scalability:** Designed the model to handle large datasets efficiently without compromising performance.

Tech Stack

- Microsoft Excel (Power Pivot, Power Query)
- DAX (Data Analysis Expressions)
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SCREENSHOTS OF THE ABOVE PROJECT:



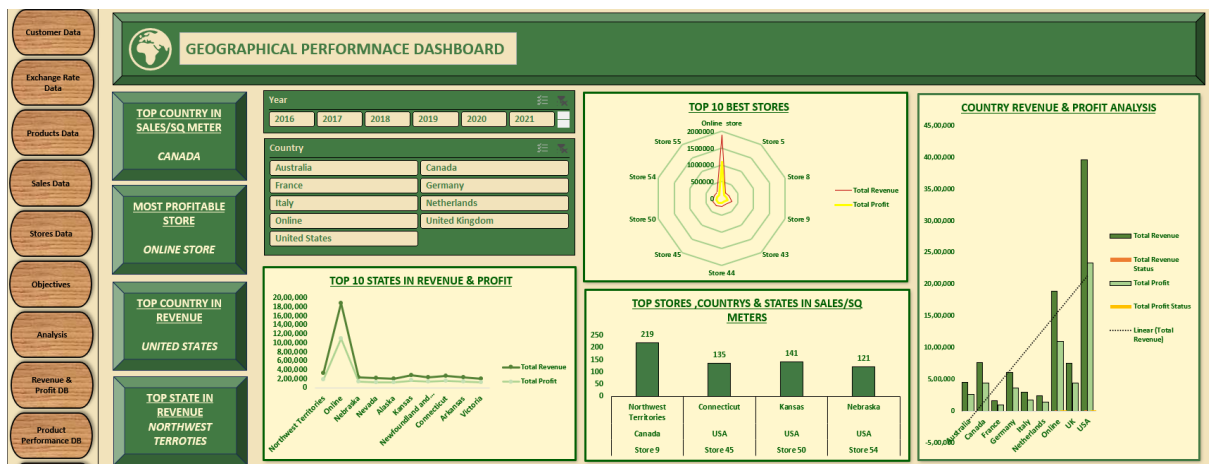
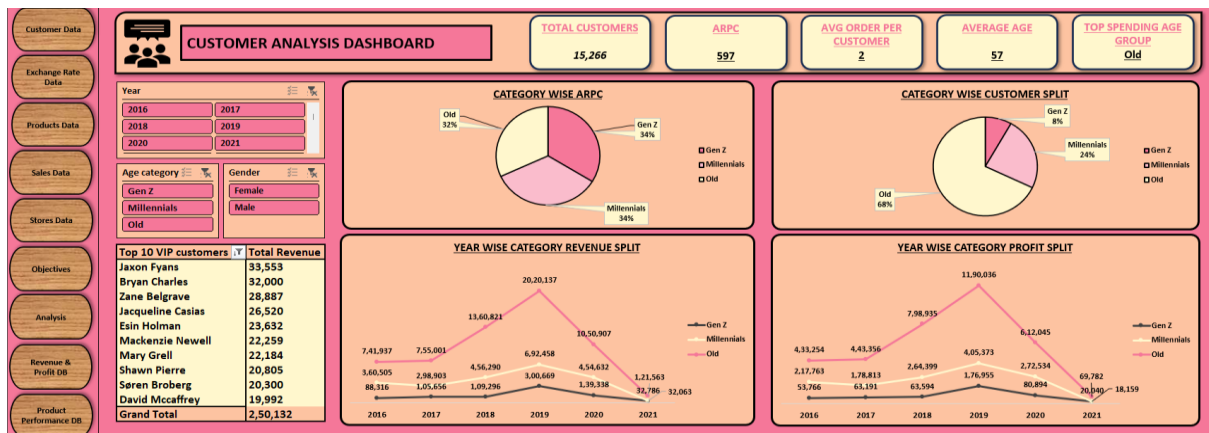
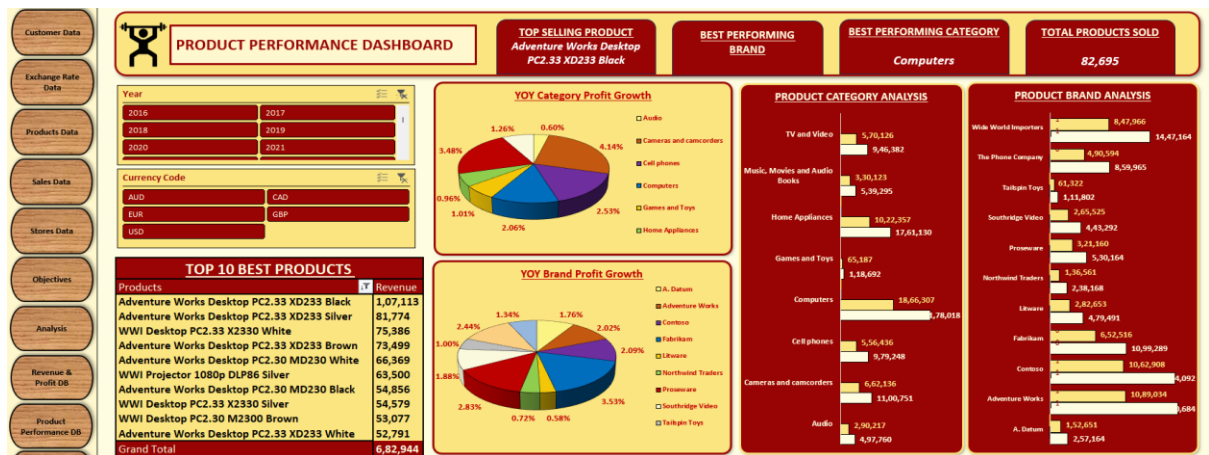
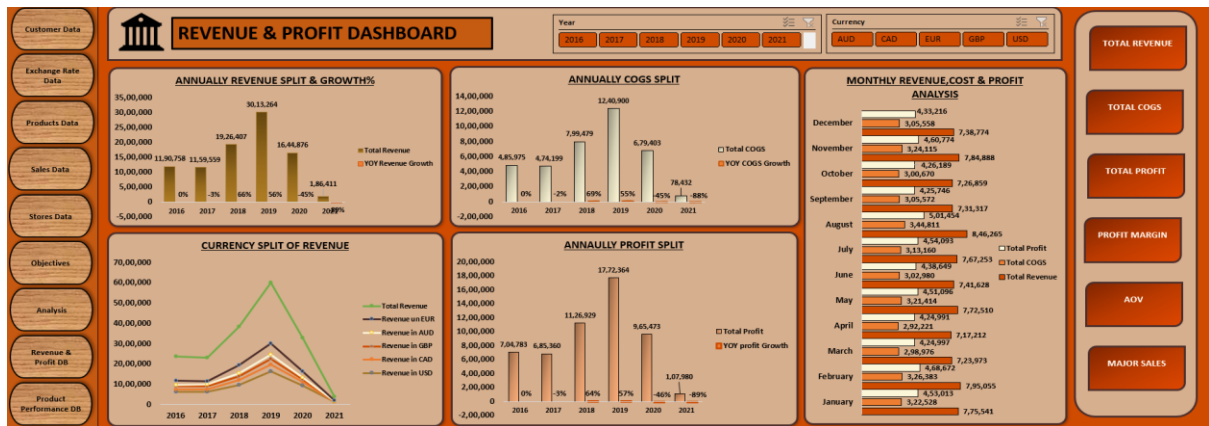
	ProductKey	Product Name	Brand	Color	Unit Cost USD	Unit Price USD	SubcategoryKey	Subcategory	CategoryKey	Category
Customer Data	1	Contoso S12MB MP3 Player E51 Silver	Contoso	Silver	6.62	12.99	101	MP4&MP3		1 Audio
	2	Contoso S12MB MP3 Player E51 Blue	Contoso	Blue	6.62	12.99	101	MP4&MP3		1 Audio
	3	Contoso 1G MP3 Player E100 White	Contoso	White	7.40	14.52	101	MP4&MP3		1 Audio
	4	Contoso 2G MP3 Player E200 Silver	Contoso	Silver	11.00	21.57	101	MP4&MP3		1 Audio
Exchange Rate Data	5	Contoso 2G MP3 Player E200 Red	Contoso	Red	11.00	21.57	101	MP4&MP3		1 Audio
	6	Contoso 2G MP3 Player E200 Black	Contoso	Black	11.00	21.57	101	MP4&MP3		1 Audio
	7	Contoso 2G MP3 Player E200 Blue	Contoso	Blue	11.00	21.57	101	MP4&MP3		1 Audio
	8	Contoso 4G MP3 Player E400 Silver	Contoso	Silver	30.58	59.99	101	MP4&MP3		1 Audio
Products Data	9	Contoso 4G MP3 Player E400 Black	Contoso	Black	30.58	59.99	101	MP4&MP3		1 Audio
	10	Contoso 4G MP3 Player E400 Green	Contoso	Green	30.58	59.99	101	MP4&MP3		1 Audio
	11	Contoso 4G MP3 Player E400 Orange	Contoso	Orange	30.58	59.99	101	MP4&MP3		1 Audio
	12	Contoso 4GB Flash MP3 Player E401 Blue	Contoso	Blue	35.72	77.68	101	MP4&MP3		1 Audio
Sales Data	13	Contoso 4GB Flash MP3 Player E401 Black	Contoso	Black	35.72	77.68	101	MP4&MP3		1 Audio
	14	Contoso 4GB Flash MP3 Player E401 Silver	Contoso	Silver	35.72	77.68	101	MP4&MP3		1 Audio
	15	Contoso 4GB Flash MP3 Player E401 White	Contoso	White	35.72	77.68	101	MP4&MP3		1 Audio
	16	Contoso 8GB Super-Slim MP3/Video Player M800 White	Contoso	White	50.56	109.95	101	MP4&MP3		1 Audio
Stores Data	17	Contoso 8GB Super-Slim MP3/Video Player M800 Red	Contoso	Red	50.56	109.95	101	MP4&MP3		1 Audio
	18	Contoso 8GB Super-Slim MP3/Video Player M800 Green	Contoso	Green	50.56	109.95	101	MP4&MP3		1 Audio
	19	Contoso 8GB Super-Slim MP3/Video Player M800 Pink	Contoso	Pink	50.56	109.95	101	MP4&MP3		1 Audio
	20	Contoso 8GB MP3 Player new model M820 Black	Contoso	Black	61.62	134.00	101	MP4&MP3		1 Audio
Objectives	21	Contoso 8GB MP3 Player new model M820 Blue	Contoso	Blue	61.62	134.00	101	MP4&MP3		1 Audio
	22	Contoso 8GB MP3 Player new model M820 Yellow	Contoso	Yellow	61.62	134.00	101	MP4&MP3		1 Audio
	23	Contoso 8GB MP3 Player new model M820 White	Contoso	White	61.62	134.00	101	MP4&MP3		1 Audio
	24	Contoso 16GB Mp5 Player M1600 Blue	Contoso	Blue	91.93	199.90	101	MP4&MP3		1 Audio
Analysis	25	Contoso 16GB Mp5 Player M1600 Black	Contoso	Black	91.93	199.90	101	MP4&MP3		1 Audio
	26	Contoso 16GB Mp5 Player M1600 Green	Contoso	Green	91.93	199.90	101	MP4&MP3		1 Audio
	27	Contoso 16GB Mp5 Player M1600 White	Contoso	White	91.93	199.90	101	MP4&MP3		1 Audio
	28	Contoso 16GB Mp5 Player M1600 Red	Contoso	Red	91.93	199.90	101	MP4&MP3		1 Audio
Revenue & Profit DB	29	Contoso 32GB Video MP3 Player M3200 White	Contoso	White	84.49	255.00	101	MP4&MP3		1 Audio
	30	Contoso 32GB Video MP3 Player M3200 Red	Contoso	Red	84.49	255.00	101	MP4&MP3		1 Audio
	31	Contoso 32GB Video MP3 Player M3200 Orange	Contoso	Orange	84.49	255.00	101	MP4&MP3		1 Audio
	32	Contoso 32GB Video MP3 Player M3200 Pink	Contoso	Pink	84.49	255.00	101	MP4&MP3		1 Audio
Product	33	Contoso 32GB Video MP3 Player M3200 Black	Contoso	Black	84.49	255.00	101	MP4&MP3		1 Audio
	34	Contoso 4GB Portable MP3 Player M450 Black	Contoso	Black	48.92	95.95	101	MP4&MP3		1 Audio
	35	Contoso 4GB Portable MP3 Player M450 White	Contoso	White	48.92	95.95	101	MP4&MP3		1 Audio

	Order Number	Line Item	Order Date	Delivery Date	Delivery status	CustomerKey	StoreKey	ProductKey	Quantity	Currency Code
Customer Data	366001	1	01-01-2016	13-01-2016	Delivered	1269051	0	1048	2	USD
Exchange Rate Data	366002	1	01-01-2016	01-12-2016	Delivered	266019	0	1108	7	CAD
Products Data	366003	1	01-01-2016		In Transit	1107461	38	163	8	GBP
Sales Data	366004	1	01-01-2016		In Transit	844003	33	421	36	EUR
Stores Data	366005	1	01-01-2016		In Transit	2035771	43	1617	1	USD
Objectives	366006	1	01-01-2016		In Transit	759705	29	1154	9	EUR
Analysis	366007	1	01-01-2016		In Transit	254340	9	1448	1	CAD
Revenue & Profit DB	366010	1	01-01-2016	01-08-2016	Delivered	370077	0	618	3	CAD
Product	366011	1	01-01-2016		In Transit	1964985	66	128	7	USD
	366012	1	01-01-2016		In Transit	1975257	65	1807	1	USD
	366013	1	01-01-2016		In Transit	1187306	41	1654	5	GBP
	366014	1	01-01-2016	01-06-2016	Delivered	738549	0	123	1	EUR
	366016	1	01-01-2016		In Transit	1562762	63	1253	7	USD
	367000	1	01-02-2016		In Transit	1438050	56	1176	1	USD
	367003	1	01-02-2016		In Transit	980164	40	153	2	GBP
	367004	1	01-02-2016		In Transit	1653699	56	136	1	USD
	367005	1	01-02-2016	01-10-2016	Delivered	754320	0	1691	1	EUR
	367006	1	01-02-2016		In Transit	693285	18	2445	3	EUR
	367007	1	01-02-2016		In Transit	1780985	57	65	7	USD
	367008	1	01-02-2016		In Transit	762346	28	1551	3	EUR
	367009	1	01-02-2016		In Transit	1512558	57	934	1	USD
	367010	1	01-02-2016		In Transit	1322939	66	917	1	USD
	367011	1	01-02-2016		In Transit	507961	22	39	1	EUR
	367012	1	01-02-2016		In Transit	209990	9	1653	3	CAD
	367013	1	01-02-2016		In Transit	815458	31	1733	2	EUR
	367014	1	01-02-2016		In Transit	1339037	38	1184	1	USD
	367015	1	01-02-2016		In Transit	911025	42	2296	7	GBP
	367016	1	01-02-2016		In Transit	1633409	63	1487	1	USD
	367017	1	01-02-2016		In Transit	1612400	47	1201	4	CAD
	367018	1	01-02-2016		In Transit	1301565	59	1245	2	USD
	367019	1	01-02-2016		In Transit	793573	29	1947	2	EUR
	367020	1	01-02-2016		In Transit	799366	29	446	6	EUR
	367021	1	01-02-2016		In Transit	243917	10	1445	4	CAD
	367022	1	01-02-2016		In Transit	942807	42	1596	1	GBP

	StoreKey	Country	State	Square Meters	Open Date
Customer Data	1	Australia	Australian Capital Territory	395	01-03-2008
Exchange Rate Data	2	Australia	Northern Territory	665	01-12-2008
Products Data	3	Australia	South Australia	2000	01-07-2012
Sales Data	4	Australia	Tasmania	2000	01-01-2010
Stores Data	5	Australia	Victoria	2000	12-09-2015
Objectives	6	Australia	Western Australia	2000	01-01-2010
Analysis	7	Canada	New Brunswick	1105	05-07-2007
Revenue & Profit DB	8	Canada	Newfoundland and Labrador	2105	01-02-2014
Product	9	Canada	Northwest Territories	1500	03-04-2005
	10	Canada	Nunavut	1210	04-04-2013
	11	Canada	Yukon	1210	06-03-2009
	12	France	Basse Normandie	350	06-06-2011
	13	France	Corse	245	06-07-2013
	14	France	Franchise-Comté	350	15-12-2009
	15	France	La Réunion	400	01-01-2013
	16	France	Limousin	385	06-03-2010
	17	France	Martinique	350	07-08-2007
	18	France	Mayotte	310	08-08-2012
	19	Germany	Berlin	1295	04-04-2015
	20	Germany	Brandenburg	1715	15-12-2012
	21	Germany	Freie Hansestadt Bremen	560	06-03-2018
	22	Germany	Freistaat Thüringen	2000	03-06-2000
	23	Germany	Hamburg	1365	01-01-2010
	24	Germany	Hessen	1855	15-12-2012
	25	Germany	Mecklenburg-Vorpommern	1610	01-01-2010
	26	Germany	Saarland	350	03-05-2015
	27	Germany	Sachsen-Anhalt	2000	08-08-2008
	28	Italy	Calabria	1200	15-12-2012
	29	Italy	Emilia	1000	01-01-2008
	30	Italy	Pesaro	2100	01-12-2008
	31	Netherlands	Drenthe	1085	01-07-2012
	32	Netherlands	Flevoland	910	01-01-2010
	33	Netherlands	Friesland	1540	12-09-2015
	34	Netherlands	Groningen	1365	01-01-2010
	35	Netherlands	Zeealand	1220	05-07-2007

	Customer Data	REVENUE AND PROFIT BASED OBJECTIVES	Product Performance Based Objective	Customer Based Objective
Exchange Rate Data	Objective 1	To analyze Year-over-Year (YoY) growth in Revenue, Profit, and COGS (%)	Objective 5	Objective 8
Products Data	Purpose:	To understand business growth trends and cost behavior over time.	To identify top-performing products based on Revenue and Profit	To analyze customer distribution by age group and its impact on revenue
Sales Data	Objective 2	To evaluate Profit Margin trends over time	Purpose:	Purpose:
Stores Data	Purpose:	To identify whether profitability is improving or declining.	To determine which products drive the majority of business value.	To identify the most valuable customer segment.
Objectives	Objective 3	To compare Revenue vs COGS to assess cost efficiency	Objective 6	Objective 9
Analysis	Purpose:	To determine how effectively the company is controlling costs.	To analyze category-wise contribution to total revenue	To compare purchasing behavior across gender
Revenue & Profit DB	Objective 4	To identify seasonal trends in monthly sales performance	Purpose:	Purpose:
Product	Purpose:	To detect peak and low demand periods.	To understand which product categories dominate sales.	To check if gender significantly influences sales.
			Objective 7	Objective 10
			To evaluate product-level profitability	To evaluate customer contribution to overall revenue
			Purpose:	Purpose:
			To identify high-revenue but low-profit products.	To understand whether revenue is concentrated among specific customer groups

	REVENUE AND PROFIT ANALYSIS	PRODUCT PERFORMANCE
Customer Data	KPI Card	KPI Card
Exchange Rate Data	Total Revenue	Top selling Product
Products Data	Revenue from every currency	Adventure Works Desktop PC2.33 XD233 Black
Sales Data	Monthly revenue Cost & profit analysis	Best performing brand
Stores Data	Yearly total revenue & growth	Adventure Works
Objectives		Best performing category
Analysis		Computers
Revenue & Profit DB		Total products sold
Product		82,695
		Product category analysis
		Category
		Audio
		Cameras and camcorders
		Cell phones
		Computers
		Games and Toys
		Home Appliances
		Music, Movies and Audio Books
		TV and Video
		Grand Total
		Top 10 Products
		WWI Projector 1080p DLP8 Silver
		Adventure Works Desktop PC2.33 XD233 Black
		Adventure Works Desktop PC2.33 XD233 Brom
		Adventure Works Desktop PC2.33 XD233 Silver
		WWI Desktop PC2.33 X2330 Silver
		WWI Desktop PC2.33 X2330 White
		Adventure Works Desktop PC2.30 MD230 Black
		Adventure Works Desktop PC2.33 XD233 White
		WWI Desktop PC2.33 X2330 Silver
		WWI Desktop PC2.33 X2330 White



	Insights & Recommendations (Objective-wise)	
Customer Data	Revenue & Profit	
Exchange Rate Data	Objective 1: YoY Growth (Revenue, Profit, COGS)	
Products Data	Current Situation Revenue and Profit show an overall increasing trend over time, but COGS is also rising alongside revenue.	
Sales Data	Insight Business is growing, but cost growth is closely following revenue growth, limiting profit expansion.	
Stores Data	Recommendation Focus on cost optimization strategies such as supplier negotiation, bulk purchasing, or operational efficiency to improve profit margins.	
Objectives	Objective 2: Profit Margin Trends	
Analysis	Current Situation Profit margins remain relatively stable with slight fluctuations across periods.	
Revenue & Profit DB	Insight The business is not significantly improving its profitability despite revenue growth.	
Product Performance DB	Recommendation Increase focus on high-margin products and reduce dependency on low-margin items.	
	Objective 3: Revenue vs COGS	Objective 5: Top Products
	Current Situation COGS increases proportionally with revenue.	Current Situation A small number of products (mainly desktop computers) generate a large share of revenue.
	Insight Indicates limited cost control efficiency — higher sales are not translating into proportionally higher profits.	Insight Business is highly dependent on a few key products.
	Recommendation Introduce cost control measures and optimize supply chain processes.	Recommendation Ensure consistent availability and promotion of top-selling products while also diversifying the product portfolio.
		Objective 6: Category Contribution
		Current Situation Categories like Computers and Home Appliances dominate total revenue.
		Insight Revenue is concentrated in specific product categories.
		Recommendation Increase focus on high-performing categories while improving strategy for underperforming ones.
		Objective 7: Product Profitability
		Current Situation Some products generate high revenue but comparatively lower profit.
		Insight Not all high-selling products are profitable.
		Recommendation Identify low-margin products and either:

THANK YOU!